The Human Cannabis Operating Manual

Chemovar, Dose, Timing, Processing, and Body-State Control

A precision guide for intentional use.

Not a prescription. Not a cure-all. Not a substitution guide.

A map of a high-variance biological system — and how to use it with enough discipline that it actually does what you intend.

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This book is not medical advice.
It does not replace consultation with a qualified healthcare provider. Cannabis interacts with medications, carries real risks, and is not appropriate for everyone. Read the Safety Wall (Part IV) before acting on anything in this guide.

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Medical disclaimer: This guide is for informational and educational purposes only. It does not constitute medical advice, diagnosis, or treatment. Always consult a qualified healthcare professional before making changes to your health regimen, especially if you take prescription medications.

Legal disclaimer: Cannabis laws vary by jurisdiction. The reader is solely responsible for compliance with applicable local, state, and federal laws.

How to Read This Guide

The Claim System

Every factual claim in this book carries an evidence grade. Read the grade before acting on the claim.

Grade	Source type	What it means
A	FDA docs, RCTs, meta-analyses, major systematic reviews	Strong clinical or regulatory evidence. The most defensible claims in the book.
B	Human trials, cohort studies, strong pharmacokinetic data	Moderate evidence. Real data, real humans, but not always large or definitive.
C	Animal studies, in vitro work, plausible mechanism	Early or mechanistic evidence. The direction is credible; human confirmation is limited.
D	Grower logic, user reports, tradition, limited validation	Field rule/anecdotal. Useful as practical guidance; not scientifically established.
X	Herby-specific methods, personal protocols, body-hack claims	Experimental or personal. Real data from a real system, not peer-reviewed.

Note

Why this matters: Cannabis research is uneven. Some things are very well established (THC psychoactivity, CBD-drug interactions, psychosis risk). Many things are early or speculative (most terpene claims, THCV effects, entourage mechanisms). This guide will not pretend otherwise. The grade is your filter.

Key Terms

Chemovar
: A cannabis variety classified by its actual chemical profile (cannabinoid and terpene content), rather than by traditional indica/sativa labels.

COA
: Certificate of Analysis. A lab report showing the measured chemical composition of a cannabis batch.

Decarboxylation
: The heat-driven conversion of acid-form cannabinoids (THCA, CBDA) into their active forms (THC, CBD).

Entourage effect
: The hypothesis that cannabinoids and terpenes interact synergistically to modify the total effect. Evidence is real but still developing. C

11-OH-THC
: 11-hydroxy-THC. The primary liver metabolite of THC after oral ingestion. More potent and longer-lasting than inhaled THC. Responsible for the stronger body load of edibles.

n-of-1 trial

A self-experiment in which the user is both the researcher and the subject. The tracking protocol in this guide is structured around this model.

Contents

Part I

Understand the Plant

CHAPTER 1

The Compound Map

What's actually in the plant, and what it may actually do.

Most cannabis conversation starts in the wrong place. It starts with the effect — *I need to sleep, I need to focus, I need to stop hurting* — and works backward to a strain name or a THC percentage. That approach fails constantly, because it skips the actual machinery.

This chapter is the machinery.

Cannabis produces hundreds of chemical compounds. A small number of them drive most of the experience. Understanding what each one is, how it behaves in the body, and how it changes with heat, time, and processing is the difference between using cannabis intentionally and just hoping for the best.

1.1 The Cannabinoids

Cannabinoids are the primary active compounds. They interact with the body's endocannabinoid system — a regulatory network involved in mood, pain, appetite, sleep, and immune response. The plant produces cannabinoids in acidic form. Heat converts them into their active form. This matters enormously for how you process and use them.

1.1.1 THC — Tetrahydrocannabinol

The primary psychoactive compound.

THC binds directly to CB1 receptors in the brain and nervous system, producing psychoactive effects, altered sensory perception, appetite stimulation, and — at the right dose — relaxation, pain modulation, and mood shift.

The dose relationship is non-linear. Low doses can produce clarity, mild euphoria, and anxiety reduction. Higher doses can flip the same person into anxiety, paranoia, and cognitive overload. This is not a defect — it is a dose-response curve that varies by individual, tolerance, food intake, sleep state, anxiety baseline, and route.

Effect	Notes	Grade
Psychoactive effects (CB1)	Dose-dependent; well established	A
Pain modulation (chronic)	Multiple trials; moderate benefit	B
Sleep onset improvement	Short-term; some rebound noted	B
Anxiety reduction (low dose)	Dose-sensitive; individual variation	B
Anxiety worsening (high dose)	Well documented; especially in naïve users	B
Nausea reduction (chemo)	FDA-approved (Marinol/dronabinol)	A

Note

Processing note: THC is the primary product of decarboxylation. Your Herby data shows this conversion actively occurring at 160 °F over 60 hours, with measurable THCA reduction and corresponding THC generation. See Chapter 2.

1.1.2 THCA — Tetrahydrocannabinolic Acid

The raw, pre-heat form of THC.

Fresh cannabis contains very little THC. It contains THCA — the acidic precursor that converts to THC when exposed to sufficient heat or UV light over time. THCA is not psychoactive in the traditional sense; it does not bind CB1 receptors the same way. But it is not inert. Early research suggests anti-inflammatory, neuroprotective, and anti-nausea properties through separate mechanisms, possibly via TRPM8 and PPARγ pathways.

Effect	Notes	Grade
Non-psychoactive vs. THC	Confirmed; lacks CB1 affinity	A
Anti-inflammatory potential	Early models only	C
Neuroprotective potential	Speculative at this stage	C

1.1.3 CBD — Cannabidiol

The most studied non-psychoactive cannabinoid.

CBD does not produce a THC-style high. Its mechanisms are broader and less direct: it modulates serotonin receptors (5-HT1A), TRPV1 channels, GPR55, and may influence how THC binds its own receptors.

The strongest clinical evidence for CBD is specific and narrow: pharmaceutical CBD (Epidiolex) reduces seizure frequency in certain rare childhood epilepsy syndromes. For the uses most people reach for — anxiety, sleep, pain, general wellness — the evidence is more preliminary and often confounded by dose variability and product quality.

One thing the clinical literature is clear on: **CBD interacts with liver enzymes.** The FDA's clinical pharmacology review of Epidiolex documents significant interactions with CYP2C19, CYP2C9, and CYP3A4. For anyone on medications that use these same pathways, CBD is not chemically neutral. See Part IV.

Effect	**Notes**	**Grade**
Seizure reduction (rare epilepsy)	FDA-approved; RCT data	A
Anxiety reduction	Multiple trials; dose matters	B
Sleep improvement	Likely via anxiety/pain reduction	B
Anti-inflammatory	Early models	C
Pain modulation	Some RCT support	B
CYP enzyme interaction	FDA-documented; clinically important	A

CBD raises blood levels of many common medications including SSRIs, warfarin, benzodiazepines, and anticonvulsants. This is not a theoretical concern — it is FDA-documented pharmacology. Full interaction data is in Part IV (Safety Wall).

1.1.4 CBDA — Cannabidiolic Acid

The raw form of CBD.

CBDA converts to CBD with heat. Some early research suggests stronger 5-HT1A binding affinity than CBD itself and potential anti-nausea activity. Most commercial products convert CBDA to CBD during processing.

Effect	Notes	Grade
Distinct from CBD	Confirmed at receptor level	B
Unique therapeutic activity	Under investigation	C

1.1.5 CBG — Cannabigerol

The parent cannabinoid. The precursor.

CBG is sometimes called the "mother cannabinoid" because CBGA is the biosynthetic precursor from which THCA, CBDA, and CBCA are all derived. High-CBG content is typically found in younger plants or specially bred chemovars.

Effect	Notes	Grade
Precursor biosynthetic role	Established biochemistry	A
Mild CB1/CB2 activity	Weak; minimal psychoactivity	C
Analgesic / anti-inflammatory	COX inhibition models	C
Anxiolytic (mice)	Animal models only	C

Note

Processing note: Herby COA data shows CBG increase after the 160 °F/60-hour processing cycle. This is consistent with known cannabinoid chemistry: as acidic precursors convert and some compounds shift, the CBG pool can change. Worth tracking across batches as a process indicator. X

1.1.6 CBN — Cannabinol

The degradation compound. What THC becomes with age and oxygen.

CBN forms when THC oxidizes — exposed to air, light, and time. It has a reputation as a sedative; the direct evidence is thin but improving. A 2024 rat study found purified CBN significantly increased both NREM and REM sleep, comparable to zolpidem. What is more reliably true: high-CBN material is old, degraded material that has also lost terpenes and potency, and the combined profile shifts toward heavier effects.

For practical purposes, CBN is a **process indicator**: rising CBN in aged or improperly stored material tells you about degradation state.

Effect	Notes	Grade
Formation from THC degradation	Established chemistry	A
Sleep increase (animal)	2024 rat study; promising	B
Sedation as isolated compound	Limited human data	C
As degradation process marker	Practical field use	D

1.1.7 THCV — Tetrahydrocannabivarin

The appetite-suppressing, shorter-duration relative.

THCV may act as a CB1 antagonist at low doses — potentially blunting THC effects and appetite — and as a partial agonist at higher doses. Early research suggests relevance to blood sugar regulation and energy metabolism. High-THCV chemovars are uncommon; African landrace genetics (Durban-type lines) have historically shown higher content.

Effect	Notes	Grade
CB1 antagonism (low dose)	Supported by receptor studies	B
Appetite suppression	Early human/animal data	C
Blood sugar / metabolic effect	Small trials; needs replication	C

1.2 The Terpenes

Terpenes are aromatic compounds produced by many plants. Cannabis produces them in the same glandular trichomes that produce cannabinoids. They are volatile — they evaporate with heat, age, and handling — making them a significant variable in processed products.

The *entourage effect* hypothesis holds that terpenes modulate cannabinoid effects through direct receptor activity, synergistic interaction, or by affecting membrane permeability. Evidence is real but still developing. Terpenes are bioactive compounds, not just flavor. B–C

1.2.1 Myrcene

Earthy, musky, herbal. The most common terpene in cannabis. Associated with sedating, relaxing profiles in practitioner literature. Direct sedation evidence at cannabis-typical concentrations is thin, but myrcene is a reasonable marker for heavier, body-oriented chemovars. C

1.2.2 β-Caryophyllene

Spicy, peppery, woody. Unusual among terpenes: it is a CB2 receptor agonist, acting on the immune and peripheral arm of the endocannabinoid system without the brain-dominant CB1 psychoactivity. Robust animal evidence for analgesic and anti-inflammatory effects. B

1.2.3 Limonene

Citrus-forward. Bright. Associated with mood elevation and anxiety reduction in early models. May modulate serotonin and adenosine receptors. Found more commonly in "sativa"-type chemovars — a correlation, not a rule. C

1.2.4 Linalool

Floral, lavender-like. Has evidence for anxiolytic and sedative activity in animal models and some human aromatherapy studies. May modulate GABA receptors. B–C (aromatherapy context stronger than ingested cannabis)

1.2.5 α-Pinene

Pine-forward. May inhibit acetylcholinesterase, potentially counteracting some THC-induced short-term memory impairment. Found in many alert-profile chemovars. Worth tracking if cognitive clarity is a target. C

1.2.6 Terpinolene

Floral, piney, herbaceous. Associated with uplifting, active profiles. Less common than myrcene or caryophyllene but often present in higher-energy chemovars. D

1.3 The Degradation Map

Compounds do not stay static. Heat, oxygen, light, and time all change the profile. This is the single most overlooked variable in most cannabis conversations.

From	To	Conditions	Grade
THCA	THC	Heat (decarboxylation)	A
CBDA	CBD	Heat (decarboxylation)	A
THC	CBN	Oxidation / aging / light	A
THC	Δ8-THC	Acid / heat / time	B
Terpenes	Evaporation	Heat, air, UV, time	A
CBGA	THC/CBD/CBC	Enzymatic (in growing plant)	A

The compound map of your raw material and your processed end product are not the same. Processing is not neutral. It is a transformation. This is why the next chapter is not a footnote — it is the second half of the foundation.

Part II

Shape the Preparation

CHAPTER 2

Processing & Chemovar Logic

How genetics, growing, harvest, cure, and transformation shape what you actually consume.

Most cannabis conversation begins with the product or preparation. It treats the preparation as a fixed object — something that either works or doesn't. This chapter argues the opposite.

The preparation is the end result of a series of decisions, each of which shifts the chemical profile in measurable ways. By the time material reaches your body, it has been shaped by the genetics of the plant, the conditions it grew in, the moment it was cut, how it was dried, how it was processed, and what it was processed in.

Understanding that pipeline does not make cannabis more complicated. It makes it more controllable.

> ***Genetics*** *are the blueprint.* ***Grow conditions*** *are the build.* ***Harvest*** *is the cut point.* ***Cure*** *is stabilization.* ***Processing*** *is transformation.* ***Route*** *is delivery.* ***Your body*** *is the final reactor.*
> *Every step modifies the output of the step before it. None of them are neutral.*

2.1 Stage 1 — Genetics: The Blueprint

Genetics determine what the plant is *capable* of producing. They encode which cannabinoid synthase enzymes are active, which terpene synthase pathways run, and what the theoretical ceiling is for THC, CBD, CBG, THCV, or any other compound. C

A plant with active THCA synthase dominance converts most of its CBGA precursor into THCA. Active CBDA synthase dominance pushes toward CBD. Plants with novel enzyme combinations produce intermediate or minority-cannabinoid-rich profiles.

What genetics do not determine: the actual expressed levels of those compounds in your final preparation. That depends on everything that comes after.

2.1.1 The indica/sativa problem

The traditional *indica/sativa* labeling system does not map reliably to chemical profile or consumer effect. It describes plant morphology and geographic origin, not chemovar. Modern cannabis genetics are extensively hybridized, and strain names are not standardized across suppliers or regions.

The stronger classification system is **chemovar** — the actual chemical fingerprint of a given batch, ideally from a verified COA. When available, use the data. When not, use terpene-dominant profile and cannabinoid ratio as the primary descriptors. B

2.2 Stage 2 — Grow Conditions: The Build

Given a genetic blueprint, cultivation conditions determine how much of that potential gets expressed.

Light: UV-B exposure is associated with upregulated terpene and THC production. Outdoor, sun-grown material often shows richer terpene profiles than equivalent indoor-grown genetics. C

Nutrients and stress: Nitrogen levels during flowering can influence metabolic priorities. Mild controlled stress (managed drought, low-stress training) can push resin production higher. Limited formal research; significant practitioner knowledge. D

Practical note: If you control or know the grow conditions of your source material, record them. If you don't, treat grow conditions as an unknown variable that explains some batch-to-batch variation.

2.3 Stage 3 — Harvest Timing: The Cut Point

Harvest timing is one of the most consequential decisions in the entire pipeline, and one of the most overlooked by end users. The same plant, harvested two weeks apart, can produce measurably different cannabinoid and terpene profiles.

2.3.1 The trichome model

Glandular trichomes change color and composition as the plant matures. This color shift is a field-practical indicator of cannabinoid state.

Trichome state	Composition	Effect tendency	Grade
Clear / transparent	Mostly THCA, minimal THC, negligible CBN	Bright, sharp, potentially racy	C
Milky / cloudy	Peak THC, low CBN	Euphoric, balanced, peak potency	C
Amber / brown	THC oxidizing to CBN, degradation products rising	Heavier, more sedating	C

The mechanism is established: THC oxidizes to CBN with heat, light, and time. Amber trichomes contain more CBN and degradation products. A 2024 rat study found purified CBN significantly increased NREM and REM sleep at doses comparable to zolpidem. B

The "clear = energy, milky = balance, amber = sedation" shorthand has real biochemical support. It is a useful field rule, not a clinical guarantee. Individual variation, genetics, and post-harvest handling all modulate the outcome. D (human effects)

2.4 Stage 4 — Cure: Stabilization

Proper curing — slow drying at 60–70 °F, 60–65 % RH, in darkness — accomplishes several things simultaneously:

- Continues slow THCA → THC conversion without significant terpene loss
- Allows chlorophyll and harsh compounds to break down
- Stabilizes terpene content against rapid evaporation

- Develops the aromatic profile as enzymatic processes complete

Poor curing — too hot, too fast, too exposed to air — accelerates terpene evaporation, forces excessive decarboxylation, and can push THC toward CBN before you have processed a single gram. C–D

2.5 Stage 5 — Processing: Transformation

This is where the source material becomes the final preparation. Processing — decarboxylation, extraction, and infusion — is where more of the final chemical profile is determined than most guides acknowledge.

2.5.1 Decarboxylation

Raw cannabis contains THCA, not THC. Heat converts acid forms to active forms through decarboxylation: the removal of a carboxyl group, releasing CO_2 and activating the compound.

The conversion is time-temperature dependent. See Appendix: Decarboxylation Reference Tables for full kinetic tables. Summary:

Temp	Conversion	Terpenes	Notes
105 °C	approx. 50 % / 90 min	Good	Slow; terpene-friendly
120 °C	80–90 % / 45 min	Moderate loss	Compromise for home use
140 °C	95–100 % / 30 min	Significant loss	Standard decarb temp
150 °C+	Fast	Major loss	Use only if speed is the sole goal

The tension is real: the temperature that converts THCA most efficiently also destroys terpenes most aggressively. There is no universally optimal decarb — only tradeoffs aligned with your target effect. C

2.5.2 Terpene boiling points

Terpene	Boiling point	Processing implication
β-Caryophyllene	130 °C (266 °F)	Lost at standard decarb temps
α-Pinene	157 °C (315 °F)	Partially survives lower-temp decarb
Myrcene	172 °C (342 °F)	Survives lower-temp; lost at high
Limonene	178 °C (352 °F)	Relatively stable at moderate temps
Linalool	198 °C (388 °F)	Most stable of the common terpenes

B (boiling point data); C (processing-to-effect implication)

2.5.3 Carrier oil selection

Cannabinoids and terpenes are lipophilic. The fat you infuse into shapes how much of the profile transfers and how long it survives storage.

A 2020 analysis (Ramella et al.) found MCT oil extracted 2–4× more terpenes than olive oil from equivalent material and maintained terpene stability over 60-day storage. Olive oil infusions lost >50 % of terpene content over the same period. Cannabinoid extraction levels were comparable between both oils initially. B

Practical rule: Use MCT or coconut oil when terpene preservation is a goal. Use olive or avocado oil for cooking integration; accept the terpene tradeoff.

Processing cannot remove medication interaction risk, impairment risk, or overuse risk. Control your preparation, but do not assume processing removes the underlying risk profile.

Herby Processing Case Study

Processing parameters: X

Temperature	160 °F (71 °C)
Duration	60 hours
Carrier oil	Coconut oil
Volume	531 mL
Oil potency	12.8 mg/mL
Total THC	6,786.1 mg

What the COA data shows:

THCA reduction Measurable decrease confirming active decarboxylation at 160 °F over extended time. This is the low-temperature, long-duration model in practice.

THC generation Corresponding THC increase as THCA converts. The 60-hour window compensates for lower temperature, achieving conversion without the aggressive heat that would maximize terpene loss.

CBG increase CBG rises after processing, consistent with known cannabinoid chemistry: as acidic precursors shift and convert, the CBG pool changes. Useful as a batch process indicator.

Terpene retention Measurable terpene retention differences consistent with the lower temperature approach. The resulting oil preserves more of the source material's terpene signature than a standard high-temp decarb would produce.

What this demonstrates: Processing parameters are not neutral. Temperature, duration, carrier, and volume all interact. The only way to build a reproducible outcome is to control and record the variables. This is not a universal recommendation — it is a documented example of one controlled protocol.

This is internal processing data from the Herby infusion system, not a certified third-party lab report. It is an operational case study of one controlled infusion protocol, not lab-certified medical proof. Processing cannot remove medication interaction risk, impairment risk, or overuse risk.

2.6 Stage 6 — Route: Delivery

How the preparation enters the body determines onset speed, duration, intensity, and which metabolic pathways are involved. Route is the final design parameter and should be decided *before* processing, not after. Full route analysis is in Chapter 3.

2.7 Stage 7 — Your Body: The Final Reactor

Same preparation. Different person. Different outcome.

The endocannabinoid system varies meaningfully between individuals. Liver enzyme activity (CYP2C9, CYP3A4, CYP2C19) determines how quickly cannabinoids are metabolized. Body composition, tolerance, sleep debt, food intake, stress, anxiety baseline, and concurrent medications all modulate the final experience.

This is why the tracking protocol later in this guide exists. You are not testing a fixed product. You are running an n-of-1 experiment in which your body is a variable, not a constant.

Part III

Use It Intentionally

CHAPTER 3

Routes & Bioavailability

How cannabis gets into the body changes what it becomes.

3.1 Why route matters

Route is the delivery layer. The same cannabinoid profile can feel very different depending on whether it is inhaled, absorbed under the tongue, swallowed, rubbed on the skin, or pushed through a patch.

That difference is not optional. It determines:

- how fast the effect arrives,
- how strong the peak feels,
- how long the experience lasts,
- how the body metabolizes the cannabinoids,
- and which risks are active.

3.2 How route changes dose

Dose is not just the milligrams on the label. Route is one of the variables that determines what that dose becomes.

If you switch from inhale to edible, the same 5 mg of THC can deliver a very different experience because the body converts part of it into 11-hydroxy-THC. That means a route change is effectively a dose change. Chapter 5 explains why route belongs in full dose planning.

3.3 Route comparison

Route	Onset	Peak	Duration	Strengths and risks
Inhaled	<10 min	10–20 min	2–4 hr	Fast control and easy titration; risk of overdoing, lung irritation, and acute anxiety with strong hits.
Sublingual	15–30 min	45–90 min	4–6 hr	Smooth onset and steadier absorption; risk of inconsistent uptake and sedation with depressant stacking.
Oral / Edible	30–120+ min	60–180 min	6–10+ hr	Long coverage and strong body load; risk of delayed overshoot, re-dosing errors, and 11-OH-THC potency.
Topical	15–60 min	Local event	1–4 hr	Local relief with minimal intoxication; risk of assum-

3.4 Inhaled

Inhaled delivery is the fastest way to feel an effect. It is best when you need a quick correction and tight feedback on dose.

- Onset is fast, which makes titration easier.
- Peak is sharp but shorter.
- It avoids the strong 11-OH-THC metabolite.

When to prefer inhaled

- acute stress interruption in users who already tolerate inhaled cannabis well,
- fast sleep onset in a short window,
- immediate pain spike control,
- and rapid focus shifts.

Inhaled red flags

- Do not use high-THC inhaled products if you are already anxious.
- Do not escalate repeatedly without pausing; a second hit can push you into a much stronger state.
- Avoid combustion if you have lung sensitivity or respiratory issues.

Route evidence grade: B

3.5 Sublingual

Sublingual oils and tinctures absorb through the mucous membranes under the tongue. This partly bypasses first-pass liver metabolism and gives a more predictable rise than oral ingestion.

- Onset is moderate.
- Duration supports several hours.
- It is a good middle ground between inhale and edible.

When to prefer sublingual

- controlled daytime effects,
- moderate anxiety support,
- sleep wind-down without a large late peak,
- and consistent low-dose protocols.

Sublingual red flags

- Absorption can be variable; let the product sit under the tongue long enough.
- Avoid stacking with benzodiazepines, opioids, alcohol, or other sedatives unless a clinician specifically approves the combination.
- Do not assume a slow onset means the dose is weak.

Route evidence grade: C

3.6 Oral / Edible / Infused Oil

Oral products are processed through the digestive system. That is the reason edibles often feel stronger than inhaled or sublingual doses.

The liver converts THC into 11-hydroxy-THC, a metabolite with a greater body load and longer duration than inhaled THC. That is why two products with the same milligrams can feel very different.

- Onset is slow.
- Peak is delayed.
- Duration is long.
- The experience often feels heavier and more body-focused.

Why edibles can feel stronger Edibles create a delayed safety feedback loop. The first signs may not appear for an hour or more, which is why premature re-dosing is the most common risk.

Oral red flags

- Wait a full 2–3 hours before evaluating the dose.
- Do not re-dose after 20–30 minutes just because you do not feel it yet.
- Use conservative doses for sleep and pain because the effect can last through the night.

Route evidence grade: B

3.7 Topical / Transdermal

Topical products are intended for local relief. They are useful when you want effect on a specific area without strong psychoactive impact.

Standard topicals usually stay local. Transdermal formulas are designed for systemic absorption.

- Standard topicals are best for surface pain and inflammation.
- Transdermal patches are best for long, steady systemic delivery.

When to prefer topical / transdermal

- targeted sore spot relief,
- recovery work after exercise,
- and times when you want reduced intoxication.

Topical / transdermal red flags

- Do not assume a topical will provide the same systemic effect as an oral or inhaled dose.
- Transdermal patches can prolong impairment, so plan for a longer window.
- Test a small skin area first to avoid irritation.

Route evidence grade: C

3.8 Route Matching by Target

Target	Preferred route	Reason
Focus	Inhale (low dose)	Fast onset; easy titration
Sleep	Oral / sublingual	Long duration; body load
Acute pain	Inhale	Rapid relief
Chronic pain	Oral	Sustained baseline coverage
Anxiety	Sublingual	Controlled onset; avoids panic spike
Recovery	Oral / topical	Duration + local option
Appetite	Oral	11-OH-THC effect enhances appetite response

[See Appendix: Pharmacokinetics Reference.]

Part IV

Safety Wall

CHAPTER 4

Medication Interactions & Red Flags

Cannabis can be useful. Cannabis can also amplify risk when stacked with the wrong body state, dose, route, or medication. Respect the system. This chapter is the guardrail.

4.1 Not medical advice / not a prescription replacement

This guide is an operational manual for product design, dosing logic, route choice, and self-tracking. It is not a medical prescription. It is not a substitute for clinician judgment.

Note

Rule: Never use this guide to remove or replace an active prescription without a qualified healthcare provider signing off.

Cannabis can interact with the same pathways as many prescription medications. When in doubt, assume an interaction exists until a clinician says otherwise. That is the safer stance.

4.2 Do not stop or reduce meds without clinician support

If you take any prescription medication, do not change your dose, stop taking it, or alter the schedule without professional guidance. That includes:

- SSRIs and SNRIs
- Benzodiazepines
- Opioids
- Anticoagulants
- Seizure medications
- Heart medications
- Psychiatric medications

Cannabis may change how your body processes these drugs. The goal is not to replace them; the goal is to avoid making the mix worse.

4.3 CYP/liver enzyme interactions

The liver enzymes that metabolize cannabinoids are the same ones that handle many pharmaceuticals. The most important ones are:

- CYP2C9
- CYP2C19
- CYP3A4
- CYP2D6

CBD and THC are not inert. They can slow or speed metabolic turnover, and that can change blood levels of co-administered drugs.

Pathway	Why it matters	Grade
CYP2C9	Metabolizes THC, warfarin, phenytoin	A
CYP2C19	Processes CBD, clobazam, omeprazole	A
CYP3A4	Handles many benzos, opioids, SSRIs	A
CYP2D6	Converts codeine, some antidepressants	B

If you take anything metabolized by these enzymes, consider cannabis an additional variable.

4.4 Prescription categories and specific risks

4.4.1 SSRIs / Lexapro

Escitalopram and similar SSRIs are common and sensitive to liver enzyme changes. CBD can inhibit CYP2C19 and may elevate SSRI blood levels. That can increase side effects such as sedation, nausea, and anxiety. A

4.4.2 Benzodiazepines

Benzodiazepines plus cannabis is additive in sedation and cognitive impairment. The combination can blunt reaction time and increase falls. Treat benzos as a serious cofactor, not a routine pairing. A

4.4.3 Opioids

Opioids and cannabis can stack sedation, slow reaction time, increase impairment, and make opioid safety harder to judge. If you take opioids, use conservative dosing only under medical oversight. A

4.4.4 Alcohol

Avoid mixing alcohol and cannabis when possible. If alcohol is already in your system, do not escalate cannabis dosing, do not drive, and do not treat your normal cannabis dose as normal. B

4.4.5 Blood thinners

Warfarin has documented concern because cannabinoid interactions may raise INR and bleeding risk. Direct oral anticoagulants such as apixaban may also be affected through CYP3A4 or P-gp pathways, but monitoring is different. If you take blood thinners, do not add high-CBD or high-THC products without clinician oversight. A

4.4.6 Seizure medications

Clobazam and other anticonvulsants are well documented to interact with CBD and cannabinoids through CYP pathways. If you take seizure medication, do not dose cannabis without a clinician who knows both. A

4.5 Body-state risks

4.5.1 Heart-rate and panic risk

High-THC products can increase heart rate and trigger panic in susceptible users. If you are anxious, sleep deprived, or under stress, start lower and choose a milder route. B

4.5.2 Psychosis / mania risk

High-THC cannabis is a risk factor for psychosis and mania in people with vulnerability. This is not a theoretical warning; it is a real risk profile. Keep THC doses conservative, especially if you have a history of mood instability. A

4.5.3 Under-25 brain risk

The under-25 brain is still developing. Frequent or high-dose THC use in this age group is associated with higher risk for cognitive, emotional, and psychiatric problems. Treat youth and young adults as a higher-risk population. A

4.5.4 Pregnancy / breastfeeding

Cannabinoids cross the placenta and appear in breast milk. The safest course during pregnancy and breastfeeding is avoidance, unless a clinician explicitly advises otherwise. A

4.5.5 Driving / workplace safety

Cannabis can impair coordination, attention, and decision-making. Do not operate vehicles, heavy equipment, or demanding machinery until you know how a dose affects you. Err on the side of extra time. A

4.6 Tolerance, dependence, withdrawal, and rebound insomnia

Cannabis is not harmless simply because it is plant-based. Frequent dosing can build tolerance, require higher doses, and trigger sleep rebound when use is reduced.

- Tolerance creep: what worked last week may feel weaker now.

- Dependence patterns: use can become the expected way to relax.
- Withdrawal / rebound insomnia: stopping suddenly can make sleep worse for a few days.

Managing these dynamics is part of responsible use. Rotate, reset, and avoid daily high-dose THC if you want the system to stay stable. **B**

4.7 Cannabis Hyperemesis Syndrome

Repeated high-dose THC use can trigger cyclic nausea and vomiting in a subset of users. This is a real syndrome, not a myth. If you develop persistent nausea, stomach pain, or repeated vomiting that improves with cannabis cessation, stop and seek medical care. **A**

4.8 Red flag checklist

If any of the following apply, stop and reassess before using again:

- You are on prescription SSRIs, benzos, opioids, blood thinners, seizure meds, or heart medication.
- You are pregnant, breastfeeding, under 25, or have a psychiatric diagnosis.
- You are planning to drive, work, or handle safety-sensitive tasks.
- You are already anxious, sleep deprived, or in a high-stress state.
- You are chasing a stronger effect after a previous dose.
- You have nausea, vomiting, or symptoms that worsen with

cannabis.

Safe cannabis use is not the absence of risk. It is the practice of recognizing and managing the risk. If you are unsure, treat the answer as "wait, record, and consult." The Safety Wall is the first part of the guide you should build your habits around.

Part V

Dosing Logic

CHAPTER 5

Dosing Logic

Dose is not just milligrams. Dose is milligrams plus route, timing, food, tolerance, medications, sleep debt, stress state, and body chemistry.

> *Dose is not just milligrams.*
> *Dose is milligrams + route + timing + food + tolerance + meds + sleep debt + stress state + body chemistry.*

5.1 The milligram is not the whole dose

It is easy to treat a cannabis dose as a single number on a label. That is the wrong model. The same 10 mg of THC can feel very different to the same person depending on:

- Route of delivery
- Food in the stomach
- Current tolerance state
- Other medications in the system
- Time of day and sleep debt
- Stress, anxiety, and body chemistry

The number on the bottle is a starting point. It is not the final dose.

5.2 Start low, go slow is engineering, not fear

Starting low and increasing slowly is not a weak strategy. It is the engineering strategy for a high-variance biological system. If you jump straight to a high number because you want a stronger result, you are betting against the system. The safer, smarter path is:

1. Choose the route.
2. Pick the lowest practical starting dose for that route.
3. Give it time to peak.
4. Record the response.
5. Adjust only one variable at a time.

If you follow those steps, you reduce the chance of surprise overdosage, panic, or a dose that lasts far longer than intended. Cross-reference Chapter 4 before you dose around medications or risk factors.

5.3 The edible delay trap

Edibles are the classic place where dose confusion becomes dangerous. Oral THC can take 45–120+ minutes to peak, and it can keep rising for hours after the first swallow. The common trap is:

- Take a dose.
- Wait 20–30 minutes.
- Assume nothing happened.
- Take more.

That is exactly the moment you create an overshoot. Wait at least 2–3 hours before deciding whether an edible dose was sufficient. The active metabolite 11-hydroxy-THC gives oral dosing a stronger body load than inhaled THC. B

Edible dosing is not faster than smoking. It is slower, longer, and often stronger. Do not re-dose before the full oral peak.

5.4 Dose stacking

Dose stacking happens when doses overlap in time. It can be deliberate or accidental.

- Inhaling, then adding an edible later the same hour.
- Taking morning oil and a second dose before the first wears off.
- Using the same route repeatedly without a reset.

Stacking can work if the goal is a stable baseline, but it also means each new dose is added to the residual effect of the previous one. That is why the adjustment strategy in this book is: change one thing at a time and track the result.

5.5 Microdose / low / medium / high categories

Dose categories are rough anchors, not hard rules. Use them to set expectations and manage risk.

Category	Typical range	Use case
Microdose	1–2 mg THC	subtle focus, mild anxiety relief
Low	2–5 mg THC	moderate relaxation, light sleep aid
Medium	5–10 mg THC	most recreational/body-load applications
High	10+ mg THC	strong sedation, deep pain relief, high tolerance

These ranges are for THC-dominant preparations. CBD, CBG, and other compounds shift the experience without changing the basic category.

5.6 Tolerance creep

Tolerance is real. If you use cannabis daily, your effective dose can move upward over time. That is not a badge of success.

- Tolerance creep reduces sensitivity.
- Higher doses can increase risk and side effects.
- Lower doses can regain power if you reset.

The practical response is to build in reset days and rotation, not to keep chasing a stronger dose. B

5.7 Reset days and taper logic

A reset day is a day when you either reduce dose substantially or skip cannabis entirely. It gives the system a chance to clear and the baseline to recover.

- For frequent users: a weekly partial reset is a useful anchor.

- For high-dose users: a taper can reduce withdrawal and rebound.
- For new protocols: start with low-dose days and only raise the frequency once the response is stable.

Taper logic means reducing the dose or spacing doses farther apart before cutting to zero. That is often easier than abrupt cessation for people using regular THC doses.

5.8 When not to dose

There are times when the safest decision is to skip cannabis. Do not dose when:

- You are taking a new medication and the interaction profile is unknown.
- You are sleep deprived, jittery, or emotionally raw.
- You have not allowed enough time for the previous dose to resolve.
- You are about to drive, work, or do anything safety-sensitive.
- You are in a high-stress or panic state.
- You are treating cannabis as a way to avoid dealing with physical or emotional issues.

5.9 One variable at a time

When you adjust your dosing, change only one variable at a time. That is the only way to know what actually changed the outcome.

- Keep route constant while changing dose.
- Keep dose constant while changing timing or food state.

- Keep your baseline routine steady while testing a new variable.

If you change multiple variables at once, you create noise instead of learning.

5.10 Dosing log template

Tracking is the bridge between intention and repeatability. Record:

- Date and time
- Route
- Dose amount and product
- Food or fasting state
- Sleep debt / energy state
- Stress or anxiety level
- Medications taken
- Outcome: onset, peak, duration, benefit, side effects
- Notes for next adjustment

> Wait. Record. Adjust. The goal is not to hit a target number. The goal is to learn how your body responds and to make each dose safer and more predictable.

Part VI

Effect Modules

CHAPTER 6

Effect-Target Modules

A practical, safety-first set of effect modules for target-driven dosing.

Each module will cover:

1. Target effect definition
2. Best-fit compound profile
3. Route recommendation
4. Dose logic
5. Avoid list
6. Evidence grade
7. Tracking notes
8. Safety warning

6.1 Sleep Module

Target: support a calm transition into sleep, sustained restorative rest, and recovery without late-night rebound.

Best-fit profile: moderate THC paired with CBD or CBN and a terpene profile that leans toward myrcene, linalool, or caryophyllene.

Route recommendation: oral or sublingual. These routes provide the duration and body load needed for sleep support and reduce the temptation to re-dose too early.

Dose logic:

- Start low and allow a full 1–2 hours for onset.
- Use smaller doses at first and increase only after tracking several nights.
- Plan for an uninterrupted sleep window; do not dose if you must wake early.
- Reserve higher nighttime doses for occasional deep-sleep sessions, not daily use.

Avoid list:

- Alcohol, benzodiazepines, opioids, or other depressants without clinician approval.
- Chasing a stronger high to force sleep.
- Late doses in the last 2 hours before morning wake time.
- Daily high-THC sleep dosing without periodic reset days.

Evidence grade: B for short-term sleep support; C for protocol design and compound targeting.

Tracking notes:

- Sleep onset time,
- number of awakenings,
- perceived sleep depth,
- next-day grogginess,
- rebound insomnia the following night.

Safety warning: Sleep dosing is not a medical sleep drug. It is a tool for building a safer bedtime routine. If you are already using prescription sleep meds or sedatives, consult a clinician before layering cannabis.

If this module conflicts with the Safety Wall, the Safety Wall wins.

6.2 Focus Module

Target: support alertness, sustained concentration, and task flow without creating jitter, overstimulation, or a false "natural Adderall" promise.

Best-fit profile: low-THC, moderate CBD or balanced ratios, with lighter terpenes such as pinene, limonene, or terpinolene.

Route recommendation: inhaled low-dose or sublingual. Inhaled is best for fast feedback and short sessions; sublingual is better when you want more stable support without a sharp peak.

Dose logic:

- Begin with a microdose or low dose.
- Adjust in small increments and test on a known task.
- Keep the route consistent while evaluating clarity versus fog.
- Avoid high-THC doses that create sedation or mental haze.

Avoid list:

- Treating this module as a stimulant replacement for ADHD medication.
- Mixing with caffeine plus cannabis in a way that masks the true effect.
- Using focus dosing during anxiety spikes or panic vulnerability.

- High-dose inhaled THC intended to force productivity.

Evidence grade: C for low-dose/task-fit logic; D / X for individualized focus protocols.

Tracking notes:

- Onset and clarity of attention,
- ability to complete a focused task,
- distractibility,
- stress level change,
- residual sedation or rebound.

Safety warning: Focus dosing is about improving task fit, not boosting performance indefinitely. If you feel anxious, restless, or mentally scattered, lower the dose and reassess.

If this module conflicts with the Safety Wall, the Safety Wall wins.

6.3 Pain Module

Target: reduce acute and chronic pain signals, support functional comfort, and provide a predictable baseline for movement and recovery.

Best-fit profile: balanced THC/CBD formulations with supportive CBG and caryophyllene-rich terpenes.

Route recommendation: oral or inhaled depending on the pain profile. Oral delivery is often best for sustained baseline coverage; inhaled low-dose delivery is useful for breakthrough spikes and tight titration.

Dose logic:

- For steady discomfort, start with a low oral or sublingual dose and observe the duration.
- For acute spikes, use a low inhaled dose and wait before adding more.
- Keep THC moderate and pair it with CBD or CBG when you want pain relief without excessive sedation.
- Track both pain reduction and functional ability, not just intensity.

Avoid list:

- Treating cannabis as a cure rather than a modulation tool.
- Using high-THC doses as an opioid replacement without clinical oversight.
- Stacking with alcohol, benzos, or opioids unless a clinician specifically approves the combination.
- Abruptly increasing dose after one underwhelming session.

Evidence grade: B for some chronic pain symptom modulation; C for individualized compound targeting and long-term protocol design.

Tracking notes:

- baseline pain score,
- response time,
- functional activity level,
- sedation or fog,
- subsequent need for rescue medication or dose reduction.

Safety warning: Cannabis can support pain management, but it is not a substitute for clinician-guided treatment. Use it as a modulation tool beside established medical care, especially if you are already on prescription analgesics.

If this module conflicts with the Safety Wall, the Safety Wall wins.

6.4 Anxiety / Stress Module

Target: support stress moderation, ease tension, and reduce acute anxiety without creating a stronger panic response.

Best-fit profile: low-THC or balanced CBD/THC products with calming terpenes such as linalool, caryophyllene, and myrcene.

Route recommendation: sublingual is preferred for many users because it offers smoother onset and more controlled entry. Inhaled low-dose delivery can work for experienced users who already tolerate inhaled cannabis and need very fast interruption.

Dose logic:

- Start with the lowest practical dose and stay there until you understand the response.
- Use a low-dose route that minimizes rapid THC spikes.
- Avoid increasing the dose during active panic or when you are already highly activated.
- If you try a new product or route, test it in a calm environment and track the response before using it under stress.
- If a dose makes anxiety worse once, treat that as data, not a challenge to overpower with more.

Avoid list:

- Avoid high-THC doses intended to force calm.
- Avoid using cannabis as a first-line treatment during an active panic attack unless you have a known, safe response.
- Avoid mixing with alcohol, benzodiazepines, opioids, or other sedatives without clinician approval.
- Avoid claiming that cannabis is a natural replacement for prescription psychiatric medications.

Evidence grade: C for low-dose stress modulation; D / X for individualized anxiety protocols and psychiatric medication overlap.

Tracking notes:

- baseline anxiety level,
- onset and strength of calm,
- any panic or overstimulation,
- perceived breathing comfort and heart-rate changes,
- next-day mood and recovery.

Safety warning: Cannabis can be part of a stress-modulation toolkit, but it is not a psychiatric medication replacement. If you are taking prescription anxiety, depression, or other psychiatric medications, consult a clinician before layering cannabis.

If this module conflicts with the Safety Wall, the Safety Wall wins.

6.5 Appetite / Nausea Module

Target: support appetite stimulation and nausea relief without treating eating disorders or serious gastrointestinal illness as a cannabis-only solution.

Best-fit profile: THC-dominant appetite support, with CBD or CBDA considered as possible nausea-modulating compounds where tolerated, plus calming terpenes such as caryophyllene, linalool, and myrcene.

Route recommendation: inhaled or sublingual can interrupt nausea quickly; oral or edible delivery is better for longer appetite support.

Dose logic:

- For nausea interruption, use a low inhaled or sublingual dose and wait before adjusting.
- For appetite support, start with a low-to-moderate oral dose and let the effect develop.
- Keep doses conservative and avoid using cannabis to override persistent or unexplained GI symptoms.
- Track both symptom relief and whether the route creates any new discomfort.

Avoid list:

- Avoid using cannabis as a substitute for medical evaluation of persistent vomiting or unexplained weight loss.
- Avoid high-THC doses solely to force hunger.
- Avoid mixing cannabis with prescription antiemetics or other GI medications without clinician oversight.
- Avoid assuming it is safe during pregnancy or chemotherapy without medical guidance.

Evidence grade: B for appetite stimulation and acute nausea support; C for protocol design in complex medical contexts.

Tracking notes:

- nausea severity,
- time to relief,
- appetite changes,
- route tolerance,
- any new GI discomfort.

Safety warning: Cannabis can support appetite and nausea, but it is not a cure for eating disorders or serious gastrointestinal conditions. Use it as a support tool alongside clinician guidance when needed.

If this module conflicts with the Safety Wall, the Safety Wall wins.

6.6 Recovery / Inflammation Module

Target: support recovery, soothe soreness perception, and help downshift inflammation-related body tension without claiming to cure injury or replace anti-inflammatory medication.

Best-fit profile: CBD/CBG-forward blends with supportive THCA and caryophyllene, plus calming terpenes such as linalool or myrcene.

Route recommendation: oral or sublingual for systemic recovery support; topical for localized soreness and inflammation perception when systemic intoxication is not desired.

Dose logic:

- Start with a low oral or sublingual dose and increase only if the response is gentle and predictable.

- Use topical applications for local soreness first, while keeping systemic doses conservative.
- Avoid using cannabis to mask injury pain, unexplained swelling, or fever.
- Track both symptom modulation and whether the dose affects mobility, comfort, and sleep.

Avoid list:

- Avoid claiming cannabis heals inflammation or repairs tissue.
- Avoid using it as a primary treatment for autoimmune disease, infection, or serious injury.
- Avoid stacking with steroids, NSAIDs, or prescription anti-inflammatory drugs without clinician approval.
- Avoid ignoring unexplained swelling, fever, or persistent pain that needs medical assessment.

Evidence grade: C for recovery-support and inflammation perception; D / X for complex autoimmune or infection-related protocols.

Tracking notes:

- soreness level,
- mobility and range of motion,
- perceived inflammation or tightness,
- sleep recovery quality,
- any fever, swelling, or new symptoms.

Safety warning: Cannabis may support recovery and soreness management, but it is not a substitute for medical care of injury, infection, or autoimmune conditions. Use it alongside clinician guidance when these issues are present.

If this module conflicts with the Safety Wall, the Safety Wall wins.

6.7 Creative / Pattern-State Module

Target: support subjective cognitive-state shifts and pattern association without promising creativity, productivity, insight, or performance enhancement.

Best-fit profile: low-to-moderate THC blends with balancing CBD or CBG, paired with terpenes such as limonene, pinene, and terpinolene to favor clarity over confusion.

Route recommendation: light inhaled or sublingual delivery for short sessions; low-dose oral delivery for longer, more deliberate pattern-state work. Avoid high-intensity concentrates and high-THC extracts when seeking manageable clarity.

Dose logic:

- Start with a low dose and wait for the full effect cycle before adjusting.
- Keep THC moderate and avoid high-THC doses that can create confusion loops or excessive dissociation.
- Choose route and dose based on session length, tolerance, and whether you can remain in a calm, supported environment.
- Track idea quality, pattern sensitivity, and sober clarity rather than chasing a creativity label.

Avoid list:

- Avoid framing cannabis as a creativity drug, productivity enhancer, or guaranteed insight generator.

- Avoid using it during mania, psychosis risk, acute panic vulnerability, or when you are responsible for driving, work, school, or important decisions.
- Avoid mixing with stimulants, sedatives, or medications that alter judgment without clinician guidance.
- Avoid treating subjective pattern-state work as a replacement for structured practice, training, or mental health care.

Evidence grade: D / X for individualized creative-state protocols; X for work, school, driving, or mental-health risk contexts.

Tracking notes:

- clarity of thought after peak,
- quality of associations and pattern sensitivity,
- confusion, dissociation, or anxiety,
- ability to resume tasks safely,
- any delayed mood or cognitive after-effects.

Safety warning: Cannabis can be a subjective pattern-state tool for some people, but it is not a guaranteed source of insight or creative performance. When in doubt, err on the side of lower dose, clearer environment, and clinician or peer support.

If this module conflicts with the Safety Wall, the Safety Wall wins.

Part VII

The Tracking System

CHAPTER 7

The Herby Protocol

Track protocols, batches, and effect sessions with the same discipline used to design them. A strong log is the final safety layer.

7.1 Herby Use Log

Use this log for every intentional session, and complete it before reflecting on the outcome.

Date / Time	
Target module	Sleep / Focus / Pain / Anxiety / Appetite / Recovery / Creative
Dose	
Route	
Food state	
Sleep debt	
Stress level	
Meds / supplements	
Onset	
Peak	
Duration	
Desired effect	
Unwanted effect	
Next-day effect	
Safety Wall conflict check	Yes / No — describe
Notes	

7.2 Herby Processing Log

Capture process details and compare batches objectively.

Batch ID	
Date	
Material source / weight	
Process type	Decarb / infusion / extraction / other
Temperature	
Time	
Carrier / solvent	
Yield	
Potency notes	
Terpene / aroma notes	
Post-process observations	

7.3 Batch Comparison Sheet

Compare how different process parameters change potency, profile, and fit for each module.

Batch	Process	Target	Outcome notes

7.4 Dose Adjustment Rules

- Start low and go slow. Change only one variable at a time.
- Use the same route when evaluating a single module, so feedback is comparable.
- If a session feels off, reduce dose or switch to a milder route

before trying a higher strength.

- Keep current sleep, stress, and food state in mind; these factors alter effect quality more than the product itself.
- Re-check the Safety Wall before every new protocol or material change.
- Document the reason for each adjustment and the observed result.

7.5 Red Flag Checklist

Stop and reassess if any of these are present before dosing.

- Mania, psychosis risk, or acute panic vulnerability.
- Driving, operating machinery, work, school, or important decisions.
- High-THC profile with poor tolerance or recent confusion.
- New or unexplained symptoms, especially pain, swelling, fever, or GI distress.
- Pregnancy, chemotherapy, or other clinically sensitive conditions.
- Medications or supplements with known CNS, cardiovascular, or metabolic interactions.
- No clear desired outcome or no agreed target module.
- Legal or situational risk in the environment.
- Lack of a sober support plan when trying a new route or dose.

7.6 Module Quick Reference

A one-line summary for each effect module.

- **Sleep:** low-dose, calming blends for bedtime transition; prioritize routine and avoid daytime impairment.
- **Focus:** balanced cannabinoid profiles for alert relaxation; avoid high-THC stimulation and late-day dosing.
- **Pain:** moderate THC/CBD support for soreness perception; avoid masking injury or stacking with contraindicated drugs.
- **Anxiety / Stress:** gentle doses with calming terpenes; avoid panic-prone routes and high-strength exposure.
- **Appetite / Nausea:** low-to-moderate oral appetite support plus possible nausea modulation; avoid using cannabis as a substitute for medical care.
- **Recovery / Inflammation:** CBD/CBG-forward recovery support and topical soreness management; avoid injury or infection replacement claims.
- **Creative / Pattern-State:** subjective cognitive-state support with clear context; avoid promising insight, productivity, or performance.

7.6.1 Route Quick Reference

- **Inhaled:** fast onset, short duration, good for quick adjustment but higher acute intensity.
- **Sublingual:** moderate onset and duration, useful for predictable effect without smoking.
- **Oral:** slow onset, long duration, good for sustained module support but requires patience.

- **Topical:** local perception management, low systemic absorption, best for soreness and recovery.

7.6.2 Evidence Grade Reference

- A: established pharmacology or consistent clinical support.
- B: moderate support from controlled studies or consistent practice reports.
- C: limited or mixed data with high individual variability.
- D: exploratory or speculative use; apply safety-first caution.
- X: not appropriate for general use outside controlled, expert-supervised context.

Decarboxylation Reference Tables

Use these temperature and time estimates as a general reference for low-temperature processing; individual material and equipment variation matters.

Temp (°C)	**THCA → THC**	**Terpene retention**	**Notes**
105	∼50 % / 90 min	Good	Slow; terpene-friendly
120	80–90 % / 45 min	Moderate light-terp loss	Home-use compromise
140	95–100 % / 30 min	Significant loss	Standard decarb
150+	Fast	Major loss	Time-critical only

Pharmacokinetics Reference

Route	Bioavail.	Onset	Peak	Duration
Inhaled (smoke/vape)	25–35 %	<10 min	30 min	2–4 hr
Oral (edible/oil)	10–20 %	30–90 min	2–4 hr	6–8+ hr
Sublingual	20–30 %	15–30 min	1–2 hr	4–6 hr
Topical (standard)	<5 %	30–60 min	Variable	Local only

All values approximate; significant individual variation exists. B–C

Evidence Grade Reference

This guide bases module claims on a relative evidence scale. Grades signal confidence and risk, not medical certainty.

- **A**: strong pharmacological or clinical evidence for the effect target.
- **B**: moderate support from controlled studies or consistent practice reports.
- **C**: limited or mixed data with high individual variability.
- **D**: exploratory or speculative support; use with safety-first caution.
- **X**: appropriate only in controlled, expert-supervised contexts or when the Safety Wall explicitly allows it.

The evidence base for this manual includes regulated cannabinoid pharmacology, terpene science, clinical review literature, and conservative safety guidance from medical and regulatory sources.

Maryland Legal Reference (2025)

Adults 21+ may possess up to 1.5 oz flower, 12 g concentrate, or 750 mg THC edibles. Home grow: up to 2 plants per person, maximum 4 per household. Consumption in private residences only. Public use remains illegal. Driving under influence is illegal.

Template for other states: see state cannabis administration websites. Possession limits, home-grow rules, and medical programme allowances vary significantly.

Terpene Quick-Reference

Terpene	Boiling pt	Aroma	Associated effect	Grade
Myrcene	172 °C	Earthy/musky	Relaxation, body load	C
β-Caryophyllene	130 °C	Spicy/clove	Anti-inflam, CB2 agonist	B
Limonene	178 °C	Citrus	Mood lift, anxiety reduction	C
Linalool	198 °C	Floral/lavender	Calming, sleep support	B
α-Pinene	157 °C	Pine	Alertness, memory support	C
Terpinolene	186 °C	Floral/pine	Uplifting, active	D
Humulene	107 °C	Earthy/woody	Anti-inflammatory	C
Geraniol	230 °C	Floral/rose	Under investigation	D